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Condiciones para el comienzo de un brote de formación estelar: estudiando el gas denso en la galaxia NGC 3110

Presentado por:
D. Pablo Martínez González

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Resumen

Este Trabajo de Fin de Grado estudia las condiciones del gas molecular en la galaxia luminosa infrarroja NGC 3110, con el objetivo de analizar su posible relación con el inicio de un brote de formación estelar y con la formación de cúmulos superestelares. Para ello, se reducen y analizan observaciones espectroscópicas obtenidas con el radiotelescopio IRAM 30 m hacia tres regiones de la galaxia: el brazo espiral norte, el centro y el brazo espiral sur. Estas observaciones incluyen las transiciones CO(1–0), CO(2–1), HCN(1–0), HCO^+(1–0) y HNC(1–0), y se complementan con mapas interferométricos de CO obtenidos con ALMA y con un mapa de densidad superficial de candidatos a cúmulos superestelares.

Las líneas CO(1–0), CO(2–1), HCN(1–0) y HCO^+(1–0) se detectan en los tres apuntados, mientras que HNC(1–0) solo se detecta de forma segura en la región central. A partir de estas líneas se derivan intensidades integradas, flujos, luminosidades de línea y cocientes moleculares. La región central domina la emisión molecular absoluta, con $L'_{\rm CO} = 2.66 \times 10^9$ K km s^{-1} pc^2, $L'_{\rm HCN} = 9.2 \times 10^7$ K km s^{-1} pc^2 y $L'_{\rm HCO^+} = 1.14 \times 10^8$ K km s^{-1} pc^2. Sin embargo, no presenta el mayor cociente $L'_{\rm HCN}/L'_{\rm CO}$. Los valores obtenidos para $L'_{\rm HCN}/L'_{\rm CO}$ son 0.040 ± 0.012, 0.034 ± 0.005 y 0.047 ± 0.008 en las regiones norte, central y sur, respectivamente, mientras que $L'_{\rm HCO^+}/L'_{\rm CO}$ toma valores de 0.042 ± 0.011, 0.041 ± 0.005 y 0.037 ± 0.008.

Estos resultados indican que NGC 3110 contiene una cantidad significativa de gas molecular asociado a trazadores de gas denso, pero no muestra un exceso extremo de luminosidad de HCN respecto a CO en línea con los resultados clásicos de Gao & Solomon (2004). Además, el cociente HCN/HCO^+ permanece próximo a la unidad en las tres regiones, por lo que no se observa una componente molecular claramente dominada por un núcleo galáctico activo. La comparación con la distribución de candidatos a cúmulos superestelares muestra que la densidad superficial de cúmulos no sigue una relación simple y monótona con los cocientes de gas denso. En conjunto, los resultados son compatibles con un escenario en el que la formación estelar intensa en NGC 3110 está favorecida por la presencia de gas molecular denso, por variaciones moderadas de excitación y por la dinámica inducida por la interacción con la galaxia compañera MCG–01–26–013. No obstante, la resolución angular de IRAM 30 m limita la identificación directa de las condiciones moleculares en las escalas propias de formación de los cúmulos superestelares.

Abstract

This Bachelor's Thesis studies the molecular gas conditions in the luminous infrared galaxy NGC 3110, with the aim of assessing their possible connection with the onset of a starburst and the formation of super star clusters. To this end, spectroscopic observations obtained with the IRAM 30 m telescope are reduced and analysed toward three regions of the galaxy: the northern spiral arm, the centre, and the southern spiral arm. These observations include the CO(1–0), CO(2–1), HCN(1–0), HCO^+(1–0), and HNC(1–0) transitions, and are complemented with ALMA interferometric CO maps and a surface-density map of super star cluster candidates.

The CO(1–0), CO(2–1), HCN(1–0), and HCO^+(1–0) lines are detected in the three pointings, whereas HNC(1–0) is securely detected only in the central region. From these lines, integrated intensities, fluxes, line luminosities, and molecular ratios are derived. The central region dominates the absolute molecular emission, with $L'_{CO} = 2.66 \times 10^9$ K km s^{-1} pc^2, $L'_{HCN} = 9.2 \times 10^7$ K km s^{-1} pc^2, and $L'_{HCO^+} = 1.14 \times 10^8$ K km s^{-1} pc^2. However, it does not show the highest L'_{HCN}/L'_{CO} ratio. The measured L'_{HCN}/L'_{CO} values are 0.040 ± 0.012, 0.034 ± 0.005, and 0.047 ± 0.008 in the northern, central, and southern regions, respectively, while L'_{HCO^+}/L'_{CO} takes values of 0.042 ± 0.011, 0.041 ± 0.005, and 0.037 ± 0.008.

These results show that NGC 3110 contains a significant amount of molecular gas associated with dense-gas tracers, but does not display an extreme excess of HCN luminosity relative to CO in accordance with the classic results of Gao & Solomon (2004). In addition, the HCN/HCO^+ ratio remains close to unity in the three regions, so no clear evidence is found for molecular emission dominated by an active galactic nucleus. The comparison with the distribution of super star cluster candidates shows that the cluster surface density does not follow a simple monotonic relation with the dense-gas ratios. Overall, the results suggest a scenario in which intense star formation in NGC 3110 is favoured by the presence of dense molecular gas, moderate excitation variations, and the dynamics induced by the interaction with the companion galaxy MCG–01–26–013. Nevertheless, the angular resolution of the IRAM 30 m observations limits the direct identification of the molecular conditions on the spatial scales where super star clusters form.

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1 Introduction

One of the central topics in astrophysics is how stars form. In this framework, molecular gas plays a fundamental role, since it constitutes the reservoir from which new stars are formed. But the molecular interstellar medium is not homogeneous, it spans a wide range of densities and physical conditions: from the bulk molecular reservoir traced by low-J CO emission to denser gas components more closely related to star-forming regions. In particular, star formation is preferentially associated with giant molecular clouds (GMCs), whose densest regions provide the conditions required for gravitational collapse.

Among the most relevant environments for studying intense star formation are systems hosting super star clusters (SSCs). These are compact and massive stellar systems, often regarded as possible young counterparts or progenitors of globular clusters. SSCs trace recent, spatially concentrated, and efficient episodes of star formation and therefore provide a direct connection between the large-scale molecular gas reservoir and extreme modes of stellar assembly. Since their formation is expected to occur in dense molecular environments, comparing CO with dense-gas tracers offers an observational route to test whether SSC-rich regions are associated with different molecular gas conditions.

1.1 Interstellar medium

The interstellar medium (ISM) is the gaseous and dusty material that fills the space between stars within galaxies. It is structured into several components that differ in density, temperature, ionisation state, and chemical composition, hence it is not homogeneous. One of its main constituents is neutral atomic hydrogen, HI, which traces the diffuse atomic medium and is most directly observed through its 21 cm hyperfine transition. This component forms part of the general gas reservoir of galaxies and provides the broader context in which the molecular phase develops.

Another major component is molecular gas, mainly composed of molecular hydrogen, H_2, with traces of other molecules such as CO, NH_3, HCN, and other species. This gas is found in molecular clouds, which have characteristic temperatures of about 10–30 K and are hence among the coldest regions of the ISM. They are detected mainly through rotational molecular lines, especially those of CO. Although molecular clouds occupy only a small fraction of the galactic volume, they contain a significant fraction of the gas mass and constitute the main sites of star formation.

The remaining volume is mainly filled by lower-density components that contribute less significantly to the total mass, such as inter-cloud gas, photo-ionised regions, and coronal gas. These phases are much hotter, with characteristic temperatures of ~ 5000, ~ 10000, and $\sim 10^6$ K, respectively, and are mainly composed of hydrogen in different ionisation states: partially ionised HI, photo-ionised HII, and collisionally ionised gas (Estalella & Anglada 1999). This multiphase structure is important because star formation is not associated with all ISM components equally.

Star formation (SF) is consequently associated with molecular clouds, although not all molecular clouds form stars with the same efficiency. According to their mass, they are commonly divided into small molecular clouds, with masses below $10^4\,M_\odot$, and giant molecular clouds (GMCs), with masses between 10^4 and $10^6\,M_\odot$. Even though this classification is somewhat approximate, observations show that massive O stars are formed systematically in GMCs rather than in lower-mass clouds. GMCs are thus regarded as the main reservoirs of molecular gas and the main sites of star formation in galaxies (Estalella & Anglada 1999).

During the star formation process, only a fraction of the available gas is eventually transformed into stars. A useful quantity to characterize this process is the star formation rate (SFR), which measures the stellar mass formed (in solar masses $M_\odot$) per unit time. Closely related to it is the star formation efficiency (SFE), which in extragalactic studies is commonly defined as

$$\mathrm{SFE} = \frac{\mathrm{SFR}}{M_{\mathrm{H_2}}}, \tag{1.1}$$

with $M_{\mathrm{H_2}}$ the molecular gas mass in $M_\odot$, and is therefore expressed in units of yr^{-1} (Domínguez-Gómez et al. 2022).

1.2 Molecular gas

Since star formation is linked to molecular gas but is not equally efficient in all molecular environments, distinguishing between the total molecular reservoir and the denser gas component is essential when studying the conditions that favour the onset of intense star formation. Although H_2 is the predominant molecule, it is difficult to observe directly under typical cold interstellar medium conditions, since it is a homonuclear diatomic molecule and hence has no permanent electric dipole moment (Estalella & Anglada 1999). For this reason, the lowest transitions of CO rotational emission is commonly used to trace the bulk molecular gas component, while high-dipole moment molecules such as HCN, HCO^+, and HNC are used as empirical tracers of denser molecular gas (Gao & Solomon 2004).

1.2.1 Molecular transitions

Molecules can emit and absorb radiation through transitions between different energy levels. In the cold molecular gas studied in this work, the most relevant transitions are rotational transitions. Their frequencies fall mostly in the millimetre and submillimetre regime of the electromagnetic spectrum. The intensity of a molecular line depends on the populations of the levels involved, which are determined by the balance between radiative and collisional processes. Hence, the excitation of a transition depends on the physical conditions of the gas, mainly its density and temperature.

A useful quantity to characterize the excitation requirements of a transition is the critical density, defined as the density at which the collisional de-excitation rate equals the spontaneous radiative decay rate:

$$n_{\mathrm{crit}} = \frac{A_{ul}}{\gamma_{ul}}, \tag{1.2}$$

where A_{ul} is the Einstein coefficient for spontaneous emission and γ_{ul} is the collisional de-excitation rate coefficient. Transitions with larger $n_{\rm crit}$ require denser gas to be efficiently excited (Estalella & Anglada 1999). For example, CO(1–0) has a critical density of $n_{\rm crit} \sim 4 \times 10^2\,{\rm cm}^{-3}$, whereas HCN(1–0), HCO^+(1–0), and HNC(1–0) have much higher critical densities of $n_{\rm crit} \sim 2 \times 10^5\,{\rm cm}^{-3}$, $3 \times 10^4\,{\rm cm}^{-3}$, and $1 \times 10^5\,{\rm cm}^{-3}$, respectively, for a kinetic temperature of $T_{\rm kin} = 30$ K. This illustrates why CO and the rest of them trace different components of molecular gas. Nevertheless, molecular emission can also be detected below the critical density, so $n_{\rm crit}$ should be understood as an approximate excitation scale rather than as a strict density threshold (Israel 2023).

1.2.2 CO as a tracer of the bulk molecular gas

In this context, CO is the most widely used tracer of molecular gas in astronomy. Its low-J rotational transitions are relatively easy to excite and are usually bright, so they mainly trace the bulk and more extended molecular reservoir.

However, the interpretation of CO emission is not completely straightforward. In many cases, the low-J transitions of CO are close to local thermodynamic equilibrium, which makes them useful for studying the physical properties of molecular clouds. Nevertheless, the most abundant isotopologue, ^{12}CO, is often optically thick, so its observed intensity does not increase linearly with the total column density[1]. Therefore, CO is a powerful tracer of the presence and distribution of molecular gas, but additional assumptions are needed when its emission is used to estimate the total molecular gas mass (Estalella & Anglada 1999).

This is commonly done through the CO-to-H_2 conversion factor, $\alpha_{\rm CO}$, which relates the CO(1–0) line luminosity to the molecular gas mass:

$$M_{\rm mol} = \alpha_{\rm CO} L'_{\rm CO(1-0)}, \tag{1.3}$$

where $L'_{\rm CO(1-0)}$ is expressed in K km s^{-1} pc^2 and $M_{\rm mol}$ usually includes the contribution from helium. A commonly adopted reference value, reviewed by Bolatto et al. (2013), is $\alpha_{\rm CO} \simeq 4.36\ M_\odot\ {\rm K}^{-1}\ {\rm km}^{-1}\ {\rm s\ pc}^{-2}$, although this factor is not universal and may vary with metallicity, gas surface density, excitation conditions, and dynamical environment.

1.2.3 Dense gas tracers: HCN, HCO^+, and HNC

While CO traces the bulk molecular reservoir, molecules such as HCN, HCO^+, and HNC provide complementary information on denser molecular gas components.

Among these molecules, HCN is one of the most widely used tracers of dense molecular gas in extragalactic studies. HCO^+ is also commonly used as a dense gas tracer, and the HCN(1–0)/HCO^+(1–0) line ratio has been used to compare the physical conditions of starburst and AGN-dominated regions. HNC, chemically related to HCN, provides complementary information on the thermal, chemical, and excitation state of the gas (Israel 2023).

[1] The number of particles of a given species integrated along the line of sight per unit area (Estalella & Anglada 1999).

1.2.4 Interpretation of molecular line ratios

The comparison between different molecular components is commonly performed through line ratios. Ratios between dense-gas tracers and CO have been widely used in extragalactic studies as empirical diagnostics of the molecular gas properties. The HCN/CO, HCO^+/CO, and HNC/CO ratios compare the emission from dense-gas tracers with the bulk molecular reservoir, while ratios between dense tracers, such as HCN/HCO^+, HNC/HCN, and HNC/HCO^+, provide additional information on excitation, chemistry, and the physical state of the gas (Gao & Solomon 2004; Israel 2023).

The HCN/HCO^+ ratio has also been used as an empirical diagnostic to distinguish between starburst- and AGN-dominated environments, since some AGN-dominated galaxy centers show enhanced HCN emission relative to HCO^+. However, this interpretation is not unique. Israel (2023) finds that HCN and HCO^+ usually have very similar intensities and trace closely related molecular gas components, while apparent HCN/HCO^+ enhancements may also reflect changes in HCO^+ emission, excitation, opacity, or chemistry rather than a genuine excess of HCN alone.

More generally, molecular line ratios must be interpreted with caution. Recent multitransition studies show that these ratios do not trace a single physical quantity in a direct way. In particular, many low-J transitions of HCN, HCO^+, and HNC can be sub-thermally excited under typical extragalactic conditions, and their observed intensities may depend on excitation, optical depth, abundance, and radiative transfer effects (Israel 2023). The HNC/HCN ratio is especially sensitive to the thermal, chemical, and excitation state of the gas, and should not be treated as a simple temperature diagnostic (Pérez-Beaupuits et al. 2007). Consequently, ratios such as HCN/CO, HCO^+/CO, HNC/CO, HCN/HCO^+, HNC/HCN, or HNC/HCO^+ are better understood as empirical indicators of the relative importance and physical conditions of different molecular gas components, rather than as unique one-to-one measurements of dense gas fraction or star formation efficiency. In this work, they are therefore used in a comparative sense, in order to investigate whether different regions of NGC 3110 show systematic changes in their molecular gas properties.

1.3 Radio astronomy

The study of cold molecular gas requires observations at radio wavelengths, particularly in the millimetre and submillimetre bands, where many rotational molecular transitions are emitted. In radio astronomy, the observed intensity, I_ν[2], is commonly expressed in temperature units through the brightness temperature, T_B. This quantity is defined by writing the specific intensity in the Rayleigh–Jeans form,

$$I_\nu = \frac{2k\nu^2}{c^2} T_\mathrm{B}, \;\Leftrightarrow\; T_\mathrm{B} = \frac{c^2}{2k\nu^2} I_\nu. \tag{1.4}$$

[2]Specific intensity, also called monochromatic intensity. It is the power received per unit frequency interval, per unit collecting area of the detector, and per unit solid angle, $\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$ (Wilson et al. 2009).

The brightness temperature is therefore a convenient way to express radiation intensity in temperature units. It coincides with the physical temperature of an optically thick thermal source only when the Rayleigh–Jeans approximation is valid; otherwise, it should be understood as an equivalent observational temperature rather than as a true thermodynamic temperature.

The interpretation of molecular emission also requires taking into account how radiation propagates through the gas. This is described by the radiative transfer equation, which includes both absorption and emission,

$$\frac{dI_\nu}{ds} = -\kappa_\nu I_\nu + j_\nu, \tag{1.5}$$

where κ_ν is the absorption coefficient and j_ν is the emission coefficient. A key quantity is the optical depth, τ, which indicates whether the medium is optically thin or thick. In the first case, the line intensity is approximately proportional to the amount of gas, while in the second one the line saturates and this proportionality is lost. Another important quantity is the source function, $S_\nu = j_\nu/\kappa_\nu$, which in local thermodynamic equilibrium is equal to the Planck function. For a homogeneous medium, the formal solution of the radiative transfer equation can be written as

$$I_\nu = I_{\nu,0} e^{-\tau_\nu} + S_\nu \left(1 - e^{-\tau_\nu}\right), \tag{1.6}$$

where $I_{\nu,0}$ is the background intensity. This expression makes explicit the connection between optical depth and observed intensity: in the optically thin limit, the observed intensity increases approximately linearly with optical depth, whereas in the optically thick limit the emission approaches the source function (Wilson et al. 2009).

Together with optical depth, the excitation state of the molecule is another key ingredient in the interpretation of spectral lines. The excitation temperature, $T_{\rm ex}$, is defined through the relative populations of the energy levels involved in the transition, according to the Boltzmann equation,

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu}{kT_{\rm ex}}\right). \tag{1.7}$$

Therefore, $T_{\rm ex}$ describes the excitation state of the molecule. If collisions are frequent enough, the excitation temperature approaches the kinetic temperature and the transition becomes thermalised. However, this condition is not always fulfilled in the interstellar medium, so many molecular lines are observed under sub-thermal excitation (Estalella & Anglada 1999).

For molecular lines, the observed line temperature depends mainly on the excitation temperature, the optical depth, and the background radiation field. In a simplified way, a line will be seen in emission when the excitation temperature is higher than the background temperature, and in absorption when it is lower. This is why the interpretation of molecular spectra is not direct: the observed intensity depends not only on the amount of gas, but also on the excitation conditions and on the radiative transfer through the medium (Wilson et al. 2009).

1.3.1 Radio telescopes and single-dish spectroscopy: IRAM 30 m

The IRAM 30 m telescope, Fig. 1, is the single-dish facility used in this work to observe the dense-gas tracers. This telescope, operated by IRAM, is located on Pico Veleta in Sierra Nevada (Granada, Spain). Its high-altitude location, at 2850 m, allows observations at frequencies between $\sim$ 80 and $\sim$ 370 GHz, while its relatively low latitude, 37° N, provides good access to the center of our Galaxy. It consists of a single-dish, alt-azimuth mounted parabolic antenna made of steel, with a total weight of about 800 tons. Its large aperture ensures high sensitivity and angular resolution, and the precision of its reflecting surface is approximately 60 μm (CDTI 2024). These characteristics make it well suited for observations of molecular gas in galaxies such as NGC 3110.

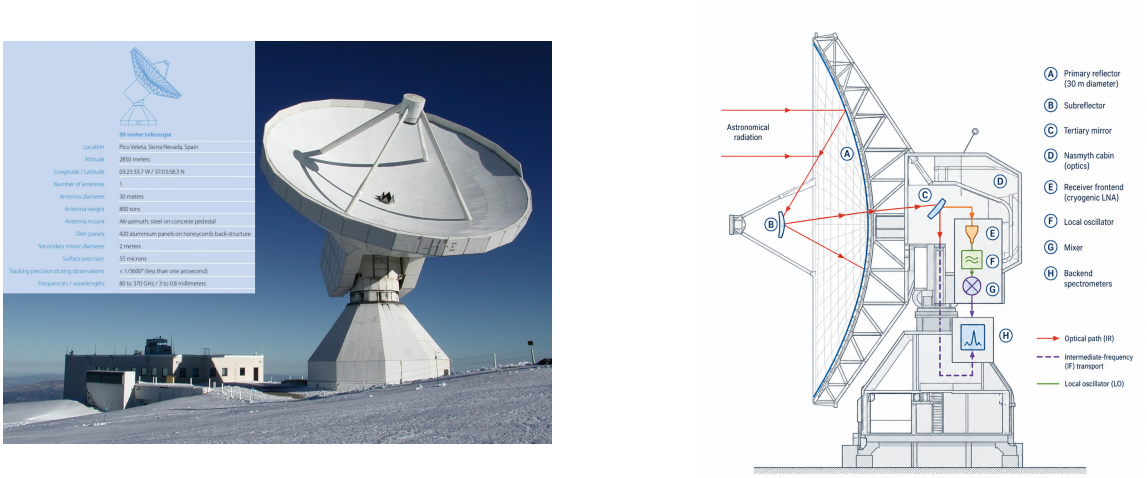


Figure 1: IRAM 30 m telescope. Left: real image of the telescope together with a summary of its main technical characteristics. Right: illustrative schematic diagram of the main components of the IRAM 30 m telescope and its signal path.

A radio telescope is an instrument designed to collect and measure electromagnetic radiation at radio wavelengths. In the case of a single-dish telescope, the incoming radiation is gathered by a large reflecting surface that concentrates the signal onto a receiver system. Unlike an ideal pencil beam, a radio telescope receives radiation from a finite range of directions on the sky. Its angular response is described by the beam pattern, whose central and most sensitive part is the main beam. The finite width of this beam defines the angular resolution of the observation and implies that the measured signal is a beam-weighted average of the source brightness distribution (Wilson et al. 2009).

From an instrumental point of view, a radio telescope is composed of several basic elements. First, the primary reflector collects the incoming radiation and redirects it toward the subreflector. In systems such as the IRAM 30 m telescope, the signal is then redirected by additional optics toward the Nasmyth cabin, where the receivers are installed. The receiver frontend amplifies and converts the astronomical signal into an intermediate-frequency signal, which is then transported to the backend. Finally, the backend analyzes the signal and produces the observed spectra. In heterodyne systems, this chain includes a local oscillator and a mixer, which are essential for converting the sky frequency into a measurable intermediate frequency (Kramer 1997).

The finite size of the beam implies that the observed signal depends not only on the intrinsic brightness of the source, but also on its angular size compared to the beam size. The observed antenna temperature can be written as the convolution of the source brightness distribution with the antenna pattern,

$$T_{\rm A} = \frac{\iint T_{\rm B}(\theta,\phi)\, P_n(\theta,\phi)\, d\Omega}{\iint P_n(\theta,\phi)\, d\Omega}, \tag{1.8}$$

where $T_{\rm B}(\theta,\phi)$ is the brightness temperature distribution of the source, $P_n(\theta,\phi)$ is the normalized antenna pattern, and θ and ϕ are angular coordinates on the sky. As a result, if the source is smaller than the beam, the measured temperature is reduced with respect to the intrinsic brightness temperature. This effect is known as beam dilution. In the opposite case, when the source fills the beam, the observed temperature approaches the source brightness temperature.

For millimetre observations from the ground, the atmosphere also plays an important role. It absorbs and emits radiation mainly due to molecules such as H_2O, O_2 and O_3, and its transmission depends strongly on frequency and on the weather conditions, particularly on the amount of precipitable water vapor. In addition, the atmosphere contributes to the total system noise, so the detected signal is not only determined by the astronomical source, but also by the atmosphere and the receiver itself. This is one of the main reasons why millimetre observations require to locate radiotelescopes at high altitudes and a careful calibration procedure before the spectra can be physically interpreted.

In a single-dish radio telescope, the astronomical signal is processed by a heterodyne receiver before being analyzed by the backend. The resulting raw spectra contain instrumental baselines that must be removed during the data reduction. Another important aspect of spectral observations is the definition of the velocity scale. Since the observed frequency of a line depends on the relative motion between the source and the observer, the spectra must be referred to a specific velocity frame. In this work, the spectra are expressed in the Local Standard of Rest velocity frame, $V_{\rm LSR}$, in order to compare the different molecular transitions consistently (Wilson et al. 2009).

At the IRAM 30 m telescope, calibration is based on the chopper wheel method, which uses observations of the sky and calibration loads to transform the backend counts into the corrected antenna temperature T_A^*. The basic idea of the method is to compare the signal of the source with the signal measured on the blank sky and with that of calibration loads of known temperature. Since the atmosphere both absorbs and emits radiation, the calibration factor depends on the atmospheric opacity, the airmass, and the response of the receiver. For this reason, the final spectrum is expressed in T_A^* units, which are already corrected for atmospheric extinction and for the forward efficiency of the telescope (Kramer 1997).

To obtain the main-beam brightness temperature, the corrected antenna temperature must be converted using the telescope efficiencies,

$$T_{\rm mb} = \frac{F_{\rm eff}}{B_{\rm eff}}\, T_A^*, \tag{1.9}$$

where $F_{\rm eff}$ is the forward efficiency and $B_{\rm eff}$ is the main-beam efficiency. The forward efficiency accounts for the fraction of the antenna response contained in the

forward hemisphere, while the main-beam efficiency measures the fraction of the detected power concentrated in the main beam. Therefore, the conversion from T_A^* to $T_{\rm mb}$ is necessary in order to express the observed signal in the main-beam brightness temperature scale (Wilson et al. 2009).

Another important parameter is the half-power beam width (HPBW), which characterizes the angular resolution of the telescope. In an ideal diffraction-limited circular aperture, the angular resolution is commonly expressed through the Rayleigh criterion as

$$\theta_{\rm diff} = \arcsin\left(1.22\,\frac{\lambda}{D}\right) \simeq 1.22\,\frac{\lambda}{D}, \tag{1.10}$$

where D is the telescope diameter, λ is the observing wavelength, and the last expression corresponds to the small-angle approximation. For the real beam of the IRAM 30 m telescope, the HPBW is usually described by the empirical approximation

$$\theta\ (\text{arcsec}) \simeq \frac{2460}{\nu\ (\text{GHz})}. \tag{1.11}$$

As a result, different molecular transitions are observed with different beam sizes and hence probe different physical scales. This must be taken into account when comparing line intensities and ratios, since part of the observed variation may arise from beam-size effects rather than from intrinsic differences in the gas properties (Kramer 1997).

1.3.2 Interferometry: ALMA

It has been proved that the angular resolution of a single dish telescope is limited, but radiotelescopes operating with the aperture synthesis technique using interferometry can improve the situation. The Atacama Large Millimeter/submillimeter Array (ALMA), operated by an international partnership between ESO[3], NRAO[4], and NAOJ[5], is located on the Chajnantor plateau in northern Chile, at an altitude of 5000 m. The exceptional atmospheric conditions at this site, characterized by very low precipitable water vapor, allow observations at millimetre and submillimetre wavelengths with minimal atmospheric absorption. ALMA consists of an interferometric array of 66 high-precision antennas, including fifty 12 m antennas and the Atacama Compact Array (ACA), which provides sensitivity to more extended spatial scales. Its interferometric nature enables angular resolutions far higher than those achievable with single-dish telescopes, making it especially suitable for studying the detailed distribution and structure of molecular gas in galaxies (ALMA Observatory 2026).

Unlike the IRAM 30 m telescope, ALMA does not measure the sky as a single-dish instrument, but combines the signal from different pairs of antennas through aperture synthesis. This technique provides high angular resolution and makes it possible to resolve compact structures within galaxies, including regions associated with intense star formation. However, interferometers are not equally sensitive to all spatial scales, and part of the extended emission may be filtered

[3]European Southern Observatory
[4]National Radio Astronomy Observatory
[5]National Astronomical Observatory of Japan

out when the corresponding short baselines are not sampled. As a consequence, interferometric maps can recover the small-scale morphology and kinematics of the gas very efficiently, but may underestimate the total flux distributed over large angular scales, a limitation known as *missing flux* (Wilson et al. 2009).

In this work, the ALMA data are used to trace the spatial distribution of the CO emission and to simulate the IRAM 30 m beam over the same regions sampled by the single-dish observations. The IRAM 30 m observations, on the other hand, provide the dense-gas spectra needed to compare the central and spiral-arm regions. The combination of both datasets makes it possible to relate dense-gas tracers and the bulk molecular gas over equivalent regions, while keeping in mind the different observational limitations of single-dish and interferometric data (Espada et al. 2018).

1.4 Our laboratory: NGC 3110

The target of this work is NGC 3110, Fig. 2, a barred spiral galaxy in the Sextans constellation and a nearby luminous infrared galaxy (LIRG). It is undergoing an early-stage interaction with the minor companion MCG–01–26–013, with a mass ratio of about 14:1. This interaction is believed to be responsible for its disturbed morphology, its asymmetric two-armed spiral structure, and its enhanced star-forming activity. Numerical simulations reproduce this morphology reasonably well and suggest that the current appearance of the system corresponds to about 150 Myr after the closest approach, during one of the highest star formation rate episodes before coalescence (Espada et al. 2018).



Figure 2: False-color RGB composite image of NGC 3110 constructed from DSS2 data, using DSS2-IR, DSS2-Red and DSS2-Blue as the red, green and blue channels, respectively. Image obtained from SkyView (NASA GSFC 2026).

This interaction-driven scenario is especially relevant because the molecular gas and star-forming activity in NGC 3110 are not uniformly distributed. Recent ALMA observations show a strong concentration of molecular gas in the central region, associated with a nuclear starburst, together with molecular gas

distributed along the spiral arms. In particular, the southern tip of the southern spiral arm shows enhanced star formation efficiency and coincides with a high surface density of SSCs, while the central region is interpreted as a site of more prolonged star formation. This suggests that, in the early stages of the interaction, the starburst activity is not restricted to the nucleus, but is also related to gas compression and fragmentation along the spiral arms (Kawana et al. 2022).

The connection between gas and cluster formation is further supported by the rich SSC population of NGC 3110. Previous studies reported more than 300 SSC candidates distributed mainly in the circumnuclear region and along the spiral arms, making this object one of the SSC-richest nearby LIRGs. In addition, Espada et al. (2018) found that regions with higher molecular gas surface densities host a larger number of SSCs, and that molecular clumps, HII regions, and SSCs are generally well co-located. This makes NGC 3110 especially suitable for investigating whether the molecular gas conditions, and in particular the balance between bulk and dense gas tracers, are related to the onset of intense star formation and SSC formation.

A complementary way to study this cluster population is through the cluster luminosity function (CLF), which is usually described by a power law and can be used to investigate how the galactic environment affects the stellar cluster population. In starburst galaxies, the CLF is often flatter than in normal spirals, suggesting a relative excess of luminous and massive clusters. In NGC 3110, the CLF also shows flatter slopes in the inner regions than in the outskirts, making it a particularly suitable case to investigate under which molecular gas conditions massive stellar clusters are formed (Espada et al. 2018).

Despite this wealth of information, an important observational gap remained. While the bulk molecular component had already been studied through interferometric CO observations by Espada et al. (2018) and Kawana et al. (2022), there were no previous published measurements of the main $J = 1 \rightarrow 0$ dense-gas tracers HCN(1–0) and HCO^+(1–0) for NGC 3110 in the literature or in public archives.

1.5 Objectives

The present Bachelor Thesis aims to fill this observational gap by reducing and analysing the IRAM 30 m observations of the dense-gas tracers HCN(1–0), HCO^+(1–0), and HNC(1–0), and by comparing them with the available interferometric CO maps from ALMA. This combination makes it possible to link the large-scale distribution of the bulk molecular gas to the denser component. In particular, the analysis is carried out in three distinct regions of NGC 3110—the central pointing and the two spiral-arm pointings—in order to explore how the molecular gas properties vary across the galaxy (Espada et al. 2022).

The specific objectives of this work are:

- To derive velocity-integrated line intensities for CO, HCN, HCO^+, and HNC in the observed regions.
- To calculate line ratios and beam-matched luminosity ratios between dense-gas and bulk molecular gas tracers.
- To compare the molecular gas properties of the center and the spiral arms.

- To discuss whether the regions with stronger dense-gas-tracer emission relative to CO are also those where the conditions are more favourable for the onset of intense star formation and the formation of SSCs.

In this way, this work seeks to assess whether the relative contribution of dense-gas tracers with respect to the bulk molecular gas in NGC 3110 is linked to the triggering of starburst activity on local scales.

2 Data

2.1 IRAM 30 m observations

The IRAM 30 m data used in this work were collected with the EMIR receiver on the 5th, 6th, and 7th of November 2022. The observations were performed in wobbler-switching mode, which allows an efficient subtraction of the sky contribution. They were configured to register the following molecular emission lines: CO (1–0), with rest frequency $\nu_{\mathrm{CO}(1-0)} = 115.2712$ GHz; CO (2–1), with rest frequency $\nu_{\mathrm{CO}(2-1)} = 230.5380$ GHz; and the dense gas emission lines HCN(1–0), HCO^+(1–0), and HNC(1–0), observed within a common 3 mm spectral setup. This setup was defined with a rest-frame equivalent central frequency of $\nu_{\mathrm{dense}} = 89.8904$ GHz, corresponding to an observed tuning frequency of $\simeq 88.4$ GHz after applying the redshift of NGC 3110.

Following the single-dish spectroscopy theory, the signal detected by the WILMA backend has been reduced using the standard IRAM 30m procedures. The adopted telescope efficiencies and parameters are shown in Table 1.

Freq. (GHz)	B_{eff}	F_{eff}	HPBW (″)
89.8904	0.80	0.95	27.83
115.2712	0.78	0.94	21.70
230.5380	0.59	0.92	10.85

Table 1: Telescope efficiencies updated to November 2016 (IRAM 2026). The Freq. column indicates the observing frequencies. B_{eff} is the main beam efficiency, F_{eff} is the forward efficiency, and HPBW is the angular resolution in arcseconds.

The data reduction was carried out using the GILDAS software (GILDAS Team 2013), provided by IRAM. The reduction process followed these main steps:

1. A baseline was subtracted from each spectrum, avoiding the velocity windows where the emission lines were expected. The baseline was fitted using a 0th-degree polynomial for HCN, HCO^+ and HNC, and a 1st-degree polynomial for the CO spectra.

2. The spectra were averaged by source, position and line, and then smoothed by merging channels.

3. The beam efficiency correction was applied to compensate for loss effects, as described previously.

4. The spectra were centered by calculating the redshifted frequency of the line and modifying it in the software.

5. The velocity window of each line was selected and fitted with a Gaussian function in order to obtain the FWHM[6], integrated intensity, and central velocity.

6. Finally, to obtain a more precise measurement, the line intensity was computed by integrating over the selected velocity channels according to Equations (2.1) and (2.2), giving the integrated intensity of each line and its associated error:

$$I = \int_{\Delta V} T_{\mathrm{mb}}(V) dV. \tag{2.1}$$

$$u(I) = \sigma \, |\Delta v| \sqrt{\frac{v_{\mathrm{max}} - v_{\mathrm{min}}}{|\Delta v|}} \tag{2.2}$$

where σ is the rms noise per spectral channel, measured outside the line window after baseline subtraction, Δv is the channel width, and $v_{\mathrm{max}} - v_{\mathrm{min}}$ is the velocity range used for the integration.

7. For the HNC observations, since no secure detection was obtained in North and South regions, an upper limit was computed using Equation (2.3):

$$I_{\mathrm{HNC}} < 3 \text{ rms } \sqrt{\Delta V \delta V}. \tag{2.3}$$

2.2 Moment 0 maps from ALMA

The ALMA analysis presented in this work is based on moment 0 maps derived from calibrated interferometric data cubes (Kawana et al. 2022). These maps correspond to the integrated intensity of the spectral lines over the selected velocity range and therefore trace the interferometrically recovered molecular gas emission along the line of sight.

The original CO (1–0) and CO (2–1) moment 0 maps are shown in Fig. 3. The CO (1–0) map is sampled on a grid of 270 × 270 pixels, with a pixel size of 0.25″ and a synthesized beam of 1.81″ × 1.43″. The CO (2–1) map is defined on a 300 × 300 pixel grid, with a pixel size of 0.30″ and a synthesized beam of 3.09″ × 2.78″. Both maps are expressed in units of Jy beam^{-1} km s^{-1}.

[6]Full Width at Half Maximum

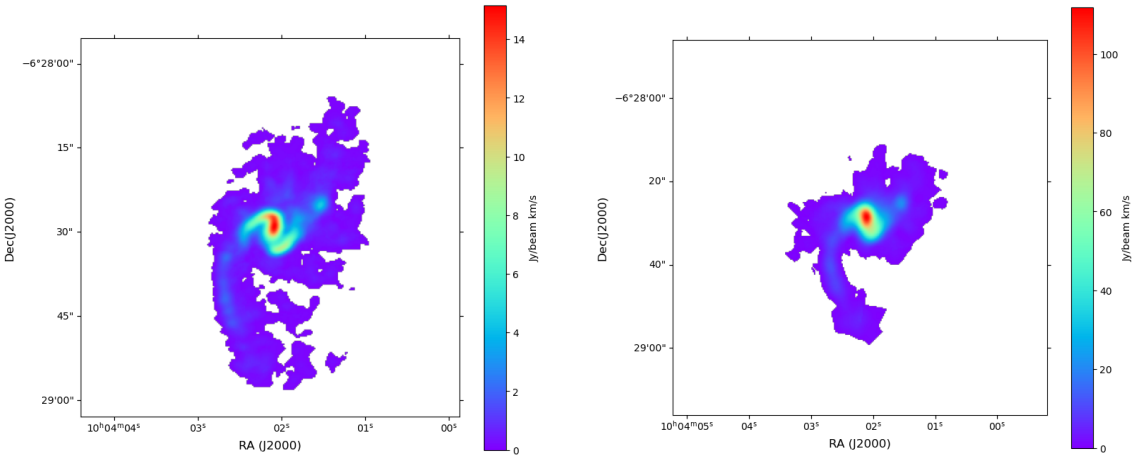


Figure 3: Original moment 0 maps of the CO (1–0) (left) and CO (2–1) (right) emission in NGC 3110. The color bar indicates the measured integrated flux range.

Since the comparison with the single-dish data requires matching the angular resolution and the regions observed with IRAM, different circular beams were considered depending on the line under study, following the HPBW values listed in Table 1. Figures 4 and 5 show the ALMA maps with the corresponding IRAM beam sizes overplotted.

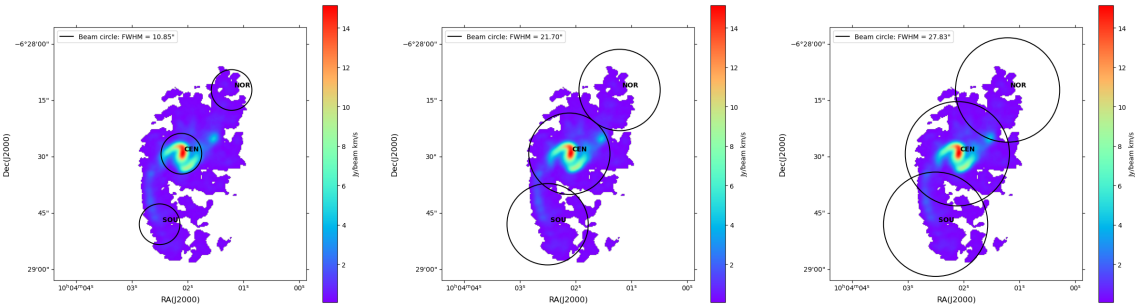


Figure 4: CO (1–0) moment 0 maps with the three IRAM 30 m pointings overlaid as circular apertures for beam sizes of 10.85″ (left), 21.70″ (center), and 27.83″ (right).

The IRAM 30 m observations were simulated directly on the ALMA moment 0 maps at each pointing. First, the ALMA maps, originally in Jy beam^{-1} km s^{-1}, were converted into Jy arcsec^{-2} km s^{-1}, by dividing by the area of the synthesized ALMA beam. A two-dimensional Gaussian kernel was then constructed for each pointing, centered at the exact sky position of the IRAM observation and adopting the corresponding circular IRAM HPBW. This kernel was normalized so that its integral over solid angle equals unity, thus reproducing the beam-averaged surface brightness that would be measured by the 30 m telescope at that position.

The final simulated flux for the IRAM 30 m beam was obtained by multiplying this surface brightness by the Gaussian kernel, yielding values in Jy km s^{-1}. In this way, the procedure reproduces the single-dish measurement directly from the interferometric map while preserving the exact spatial weighting of the IRAM beam.

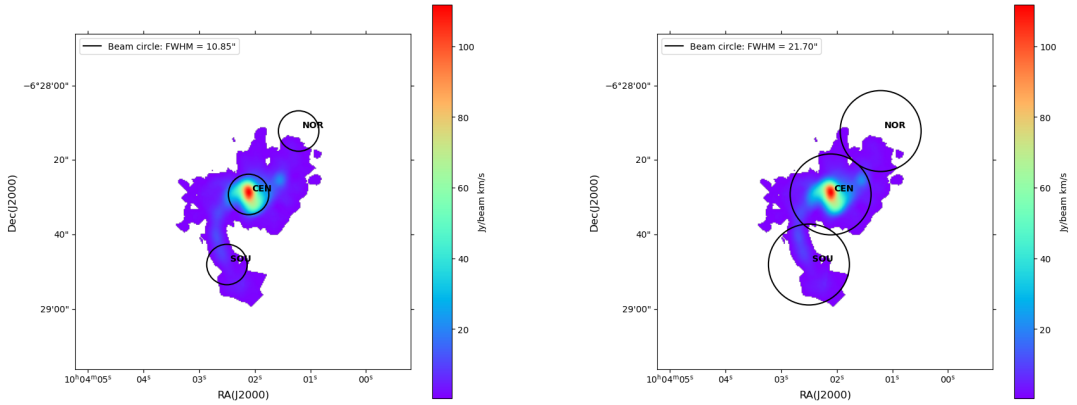


Figure 5: CO (2–1) moment 0 maps with the three IRAM 30 m pointings overlaid as circular apertures for beam sizes of 10.85″ (left) and 21.70″ (right).

This procedure was applied to the original CO (1–0) map using the IRAM beams of CO (1–0), CO (2–1), and the dense gas observations, and to the CO (2–1) map using the IRAM beams of CO (2–1) and CO (1–0). This makes it possible to compare fluxes and line ratios over equivalent physical scales.

Due to the missing flux problem, the ALMA-based simulated fluxes may still underestimate the total emission with respect to the single-dish IRAM measurements, CO(1–0) recovered flux was 77% ±15% compared to single-dish measurements (Kawana et al. 2022). In addition, the masking procedure applied when constructing the interferometric moment 0 maps may exclude part of the faint or extended emission if it falls below the adopted selection thresholds. This effect may be particularly relevant in the North and South pointings, where the IRAM beams encompass regions with weak or absent recovered CO(2–1) emission in Fig. 5. Taking all of this into account, the simulated ALMA fluxes in the spiral arms should be interpreted with caution, since both interferometric filtering and masking effects may contribute to an underestimation of the total CO(2–1) emission.

2.3 SSC map

Since one of the main objectives of this work is to investigate the molecular gas conditions associated with starburst triggering, the spatial distribution of super star cluster candidates in NGC 3110 provides a useful reference to compare with the molecular line ratios derived from the IRAM 30 m observations.

NGC 3110 was observed in the near-infrared K_s band with VLT/NACO as part of the SUNBIRD survey (Randriamanakoto 2015; Randriamanakoto & Väisänen 2017). The corresponding image is shown in Fig. 6 and is used as the morphological reference for the SSC candidate population identified in previous works (Espada et al. 2018).

The SSC surface-density map used in this work is shown in Fig. 7. This map represents the projected number density of the K_s-selected SSC candidates. Espada et al. (2018) reported a large population of SSCs in NGC 3110, mostly located along the spiral arms and in the circumnuclear region, and found that regions with higher molecular gas surface density tend to host a larger number of SSCs.

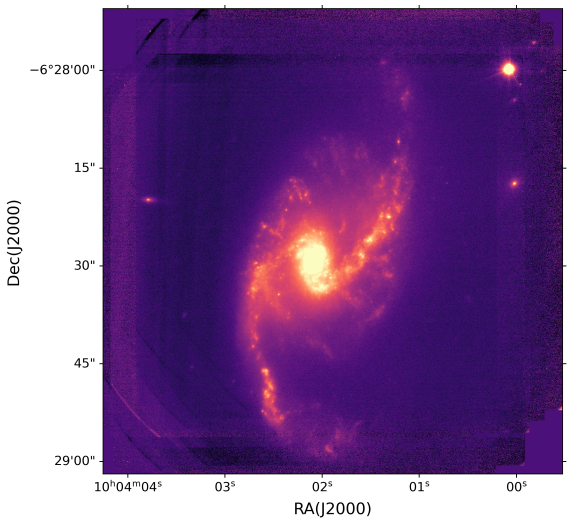


Figure 6: Near-infrared K_s-band image of NGC 3110 obtained with VLT/NACO. This image provides the morphological reference from which the SSC candidate population was identified (Randriamanakoto 2015; Espada et al. 2018).

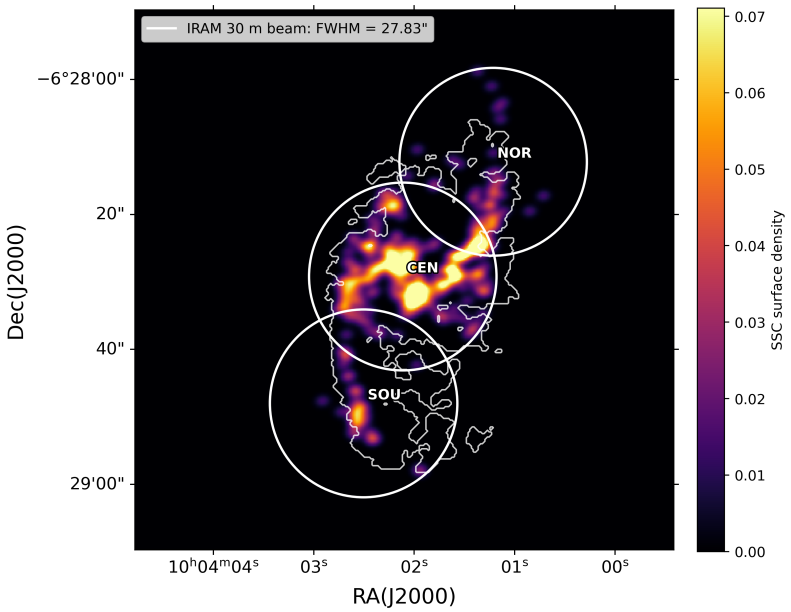


Figure 7: Surface-density map of the K_s-selected SSC candidates in NGC 3110. The map represents the projected number density of compact SSC candidates. White circles indicate the three IRAM 30 m pointings analyzed in this work, with a diameter corresponding to the 27.83″ beam of the dense-gas observations. The white contour shows the CO(1–0) molecular-gas mask used to define the area over which the SSC surface density is normalized.

The SSC population sampled by each IRAM 30 m pointing is quantified by counting the candidates enclosed within the same circular apertures used for the dense-gas observations. In order to define a surface density that can be compared with the molecular gas properties, the normalization area is defined from the CO(1–0) molecular-gas mask inside each aperture. For each pointing i, the adopted molecular area is computed as

$$A_{\mathrm{CO},i} = N_{\mathrm{CO},i}, A_{\mathrm{pix}}, \tag{2.4}$$

where $N_{\mathrm{CO},i}$ is the number of CO(1–0)-masked pixels inside the i-th IRAM 30 m aperture and A_{pix} is the physical area of one pixel. The SSC surface density is then computed as

$$\Sigma_{\mathrm{SSC},i} = \frac{N_{\mathrm{SSC},i}}{A_{\mathrm{CO},i}}. \tag{2.5}$$

This definition measures the number of clusters per unit area occupied by molecular gas within each IRAM pointing. This is the relevant quantity for comparing the SSC distribution with the dense-to-bulk molecular gas ratios derived in the same regions.

The resulting values are listed in Table 2. The central pointing contains both the largest number of SSC candidates and the highest CO-normalized SSC surface density. This is consistent with the strong circumnuclear concentration of SSCs reported in previous studies (Espada et al. 2018; Kawana et al. 2022). The northern aperture shows an intermediate value while the southern pointing gives the lowest surface density. Although the southern aperture contains fewer SSC candidates than the northern one, its CO-covered area is larger, which lowers the number of SSCs per unit molecular-gas area.

	North	Center	South
N_{SSC}	66	274	47
A_{CO} [kpc^2]	20.38	51.05	28.80
f_{CO}	0.32	0.79	0.45
Σ_{SSC} [kpc^{-2}]	3.24	5.37	1.63

Table 2: Area-normalized SSC surface density within each IRAM 30 m dense-gas pointing. The normalization area A_{CO} is defined from the CO(1–0) molecular-gas mask inside each circular aperture. f_{CO} gives the fraction of the aperture covered by the CO(1–0) mask.

3 Results

Once the data reduction from IRAM observations, the simulations over the ALMA maps and the SSCs surface density are complete, the results are presented below.

3.1 Spectra from IRAM 30 m

The IRAM 30 m spectra shown in Figs. 8 and 9 provide the observational basis for the integrated intensities and line ratios derived in the following subsection. They allow a direct comparison of the detectability, peak intensity, line width, and velocity centroid of each molecular transition at the three observed positions in NGC 3110.

The two CO transitions are clearly detected in the North, Center, and South pointings, and show the broadest and strongest profiles in the sample, as expected for tracers of the bulk molecular gas reservoir. Their widths are in accordance with the large velocity range previously reported for NGC 3110 from interferometric CO(2–1) observations, where emission is detected between 4780 and 5240 km s^{-1} in the radio LSRK frame (Espada et al. 2018). In contrast, the dense gas tracers are significantly weaker. HCN(1–0) and HCO$^+$(1–0) are detected in the three pointings, whereas HNC(1–0) is only securely detected in the central position, with upper limits derived for the North and South spectra.

For the CO transitions, the velocity axis is expressed relative to the systemic velocity of NGC 3110, adopting $v_{\rm sys} = 4990$ km s^{-1} in the radio definition and LSRK frame (Espada et al. 2018). In this convention, the plotted CO velocity is defined as

$$v_{\rm CO,rel} = v_{\rm CO,LSRK} - v_{\rm sys}. \tag{3.1}$$

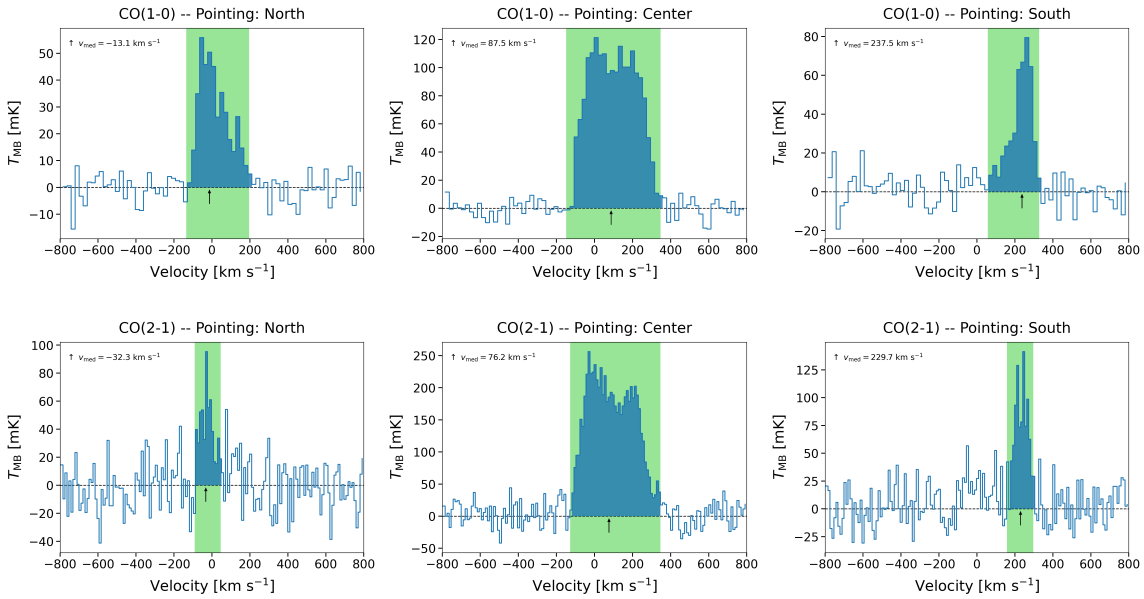


Figure 8: IRAM 30 m spectra of the CO(1–0) and CO(2–1) transitions toward the North, Center, and South pointings in NGC 3110. The velocity axis is expressed relative to the systemic velocity of the galaxy, adopting $v_{\rm sys} = 4990$ km s^{-1} in the radio definition and LSRK frame (Espada et al. 2018). The vertical arrows mark the residual centroid velocities $v_{\rm med}$ obtained from Gaussian fits. Green shaded regions indicate the velocity intervals used to compute the integrated intensities and the filled blue areas show the emission included in the integration.

The arrows in Fig. 8 mark the residual centroid velocities obtained from the Gaussian fits. These values do not represent an error in the reduction, but the beam-averaged line-of-sight velocity of each pointing with respect to the systemic velocity of the galaxy. Hence, the fact that the CO centroids do not coincide exactly with $v = 0$ reflects the local kinematics sampled by the IRAM 30 m beam in the nucleus and spiral-arm regions.

For the dense gas tracers, the velocity axis is defined independently for each molecular transition. In each spectrum, the reference frequency is the redshifted frequency expected for that particular line using the systemic reference adopted for NGC 3110. Therefore, $v = 0$ corresponds to the observed frequency expected for HCN(1–0), HCO^+(1–0), or HNC(1–0) after applying this reference redshift.

For a transition with rest frequency ν_0, the reference frequency is

$$\nu_z = \frac{\nu_0}{1 + z_{sys}}, \tag{3.2}$$

where $z_{sys} = 0.01664$[7] is the reference redshift associated with the adopted systemic velocity of NGC 3110. The velocity axis shown in Fig. 9 is then measured with respect to this redshifted line frequency. In this convention, a line centered at $v \simeq 0$ would be consistent with the adopted systemic reference, while a non-zero centroid may reflect local gas kinematics within the corresponding pointing, together with small differences introduced by the adopted velocity convention and reference-frame treatment.

A visual inspection of the spectra reveals clear differences between the three observed regions. The central pointing shows the strongest dense gas emission, with HCN(1–0), HCO^+(1–0), and HNC(1–0) all detected within a similar velocity range. This indicates that the central region contains the most prominent dense molecular gas component among the three IRAM 30 m pointings. In contrast, the North and South pointings show fainter dense gas emission. HCN(1–0) and HCO^+(1–0) are still detected in both spiral-arm positions, while HNC(1–0) is not securely detected there and is treated through upper limits.

The fitted centroids of the detected lines are generally shifted with respect to the systemic reference of each transition. In a purely rotating disk, opposite sides of the galaxy would be expected to show velocity offsets with opposite signs; consequently, the fact that the northern and southern pointings do not simply follow this pattern suggests that local non-circular motions are also present. These shifts are thus more likely related to regional kinematic differences within the disturbed molecular disk, such as streaming motions along the spiral arms or interaction-driven perturbations, rather than to systematic offsets between the molecular transitions (Kawana et al. 2022).

The spectra are also useful for assessing the robustness of the adopted integration windows. The shaded green regions indicate the velocity intervals used to compute the integrated intensities, while the filled areas correspond to the emission included in the integration. For HNC(1–0) in the North and South pointings, where no secure detection is obtained, the red rectangles illustrate the velocity width and amplitude adopted to estimate the corresponding upper limits.

[7] $z_{sys} = v_{sys}/c$, with $v_{sys} = 4990$ km,s^{-1} following Espada et al. (2018).

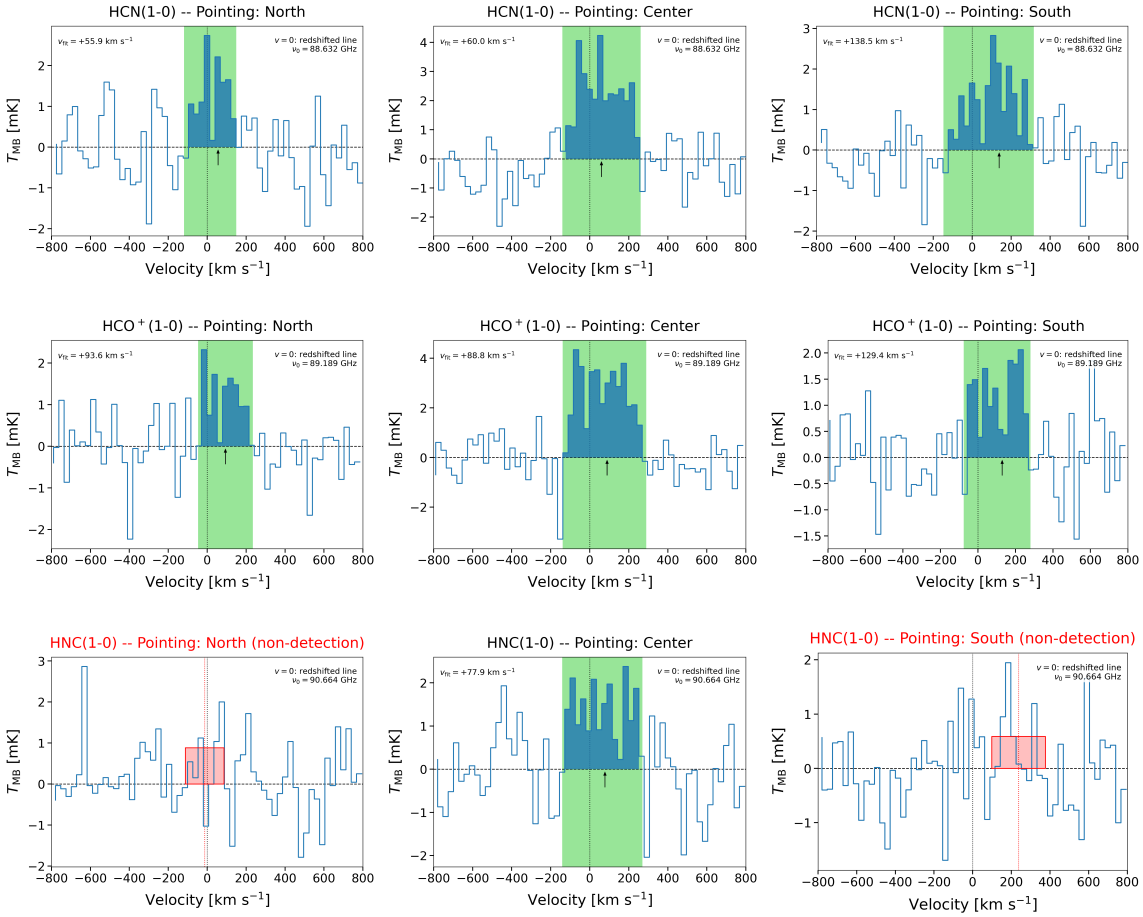


Figure 9: IRAM 30 m spectra of the dense gas tracers HCN(1–0), HCO^+(1–0), and HNC(1–0) toward the North, Center, and South pointings in NGC 3110. For each transition, the velocity axis is centered on the redshifted frequency expected for that specific line. The vertical dotted line shows this reference, while the black arrow marks the Gaussian centroid for detected dense gas lines. Green shaded regions indicate the integration windows, blue filled areas show the integrated emission, and red rectangles mark the upper-limit estimates for HNC non-detections.

Overall, Fig. 9 supports the use of HCN(1–0) and HCO^+(1–0) as the most reliable dense gas detections in the three pointings, while HNC(1–0) provides useful additional information mainly in the central region. This behavior is consistent with the lower signal-to-noise expected for dense gas tracers in extragalactic single-dish observations and with the fact that HCN, HCO^+, and HNC trace denser molecular material than low-J CO, although their intensities should not be interpreted as direct dense gas masses without considering excitation and chemical effects (Jiménez-Donaire et al. 2019).

3.2 Molecular gas magnitudes and ratios

The main quantities derived from the observations are the integrated line intensities measured from the IRAM 30 m spectra. Table 3 lists the velocity-integrated intensities obtained for each molecular transition and pointing, as well as the Half Power Beam Width (HPBW) of the telescope beam, which depends on the observing frequency.

Line	HPBW (arcsec)	North (K km s^{-1})	Center (K km s^{-1})	South (K km s^{-1})
CO(1–0)	21.70	9.0 ± 0.5	38.8 ± 0.6	8.2 ± 0.4
CO(2–1)	10.85	5.5 ± 0.7	69.9 ± 1.3	10.0 ± 0.8
HCN(1–0)	27.83	0.32 ± 0.10	0.85 ± 0.11	0.50 ± 0.08
HCO^+(1–0)	27.83	0.32 ± 0.09	1.05 ± 0.10	0.40 ± 0.08
HNC(1–0)	27.83	< 0.18	0.49 ± 0.12	< 0.16

Table 3: Integrated line intensities measured with the IRAM 30 m telescope toward the three pointings in NGC 3110. For each molecular transition, the table lists the half-power beam width (HPBW) and the velocity-integrated intensity (I) measured in the North, Center, and South positions.

An additional relevant quantity is the integrated flux derived from the ALMA CO(1–0) and CO(2–1) moment 0 maps. These values are presented in Table 4. In that table, the Full Width at Half Maximum (FWHM) indicates the width of the two-dimensional Gaussian kernel applied to the ALMA maps in order to simulate the corresponding IRAM 30 m beam.

Line	FWHM (arcsec)	North (Jy km s^{-1})	Center (Jy km s^{-1})	South (Jy km s^{-1})
CO(1–0)	10.85	5.1 ± 0.3	129 ± 7	14.2 ± 0.7
	21.70	31.0 ± 1.6	187 ± 9	50 ± 3
	27.83	64 ± 3	206 ± 10	88 ± 4
CO(2–1)	10.85	4.5 ± 0.8	354 ± 35	26 ± 3
	21.70	59 ± 6	494 ± 49	106 ± 11

Table 4: Integrated fluxes measured from the ALMA maps of NGC 3110. For each transition, the table lists the Gaussian FWHM adopted for the measurement and the integrated flux (S) in the North, Center, and South regions.

The uncertainty associated with each beam-simulated flux was estimated by propagating a constant rms per pixel through the Gaussian beam weighting. In this way, the statistical error reflects how the uniform map noise contributes to the simulated pointing once convolved with the IRAM 30 m beam. Moreover, systematic calibration uncertainties of 5% for the CO(1–0) maps and 10% for the CO(2–1) maps were added in quadrature to the statistical uncertainty.

To compare the beam-simulated ALMA fluxes in Table 4 with the IRAM integrated intensities in Table 3, both quantities are expressed in consistent units. Using the Rayleigh–Jeans relation, the conversion between velocity-integrated main-beam temperature, $I_{\rm line} = \int T_{\rm mb}\, dv$, and integrated flux density, $S_{\rm line}\Delta v = \int S_\nu\, dv$,

for a Gaussian beam is

$$S_{\rm line}\Delta v = \frac{\nu^2\,\theta_{\rm maj}\theta_{\rm min}}{1.222\times10^6}\,I_{\rm line} \Leftrightarrow I_{\rm line} = 1.222\times10^6\,\frac{S_{\rm line}\Delta v}{\nu^2\,\theta_{\rm maj}\theta_{\rm min}}, \tag{3.3}$$

where $S_{\rm line}\Delta v$ is in Jy km s^{-1}, $I_{\rm line}$ in K km s^{-1}, ν in GHz, and $\theta_{\rm maj}$ and $\theta_{\rm min}$ are the FWHM axes of the beam in arcsec. For the IRAM 30 m spectra, Eq. (3.3) can be evaluated using the HPBW–frequency relation given in Eq. (1.11), with a circular beam $\theta_{\rm maj} = \theta_{\rm min} = \theta$ (Kramer 1997). This gives

$$S_{\rm line}\Delta v \simeq \frac{2460^2}{1.222\times10^6}\,I_{\rm line} \simeq 4.95\,I_{\rm line}. \tag{3.4}$$

This is used to transform the IRAM CO intensities into fluxes and compare them with the beam-simulated ALMA measurements, allowing the flux recovery of the interferometric maps to be quantified. The inverse relation is used when the ALMA fluxes are expressed as beam-matched integrated intensities. After matching the angular resolution of the two CO transitions, the ratio $I_{\rm CO(2-1)}\,/\,I_{\rm CO(1-0)}$ is interpreted in temperature units as an excitation diagnostic: values close to unity are expected for optically thick and thermalised gas in the Rayleigh–Jeans limit, while lower values indicate colder, lower-density, or sub-thermally excited molecular gas. Values above unity may suggest warmer or more compact emission, but can also be affected by missing flux and calibration uncertainties (Domínguez-Gómez et al. 2022).

Table 5 presents the ratio between the IRAM and ALMA fluxes for the same CO transition. As already suggested by looking at Fig. 5 (left), the CO(2–1) map simulated with the IRAM beam at $\nu_{\rm CO(2-1)}$ does not recover the entire relevant emission resulting in a large difference between measurements, which is especially severe in the North region but still significant in the South. A value sufficiently close to unity (0.95 ± 0.10) suggests that, in the central pointing, the mismatch between IRAM and ALMA is less severe than in the outer regions.

Ratio	HPBW (arcsec)	North	Center	South
CO(1–0) IRAM/ALMA	21.70	1.20 ± 0.09	0.86 ± 0.05	0.68 ± 0.05
CO(2–1) IRAM/ALMA	10.85	5.9 ± 1.2	0.95 ± 0.10	1.9 ± 0.3

Table 5: Comparison between IRAM and ALMA integrated fluxes in NGC 3110. The listed values correspond to the IRAM/ALMA beam-matched integrated-flux (S) ratio in the North, Center, and South positions. The HPBW column indicates the common angular resolution adopted for each transition.

Table 6 shows the CO(2–1)/CO(1–0) ratios obtained after matching both transitions to a common angular resolution. The first row compares the ALMA CO(2–1) emission with the IRAM CO(1–0) measurement at 21.70″, while the second row combines the IRAM CO(2–1) intensity with the ALMA CO(1–0) map simulated at 10.85″. The Center values are mutually compatible within the uncertainties, whereas the North and South pointings show stronger variations with angular resolution. These differences are discussed in Section 4, where the possible effects of excitation, compact emission, and missing flux are considered.

Ratio	Order	HPBW	North	Center	South
$\frac{CO(2-1)}{CO(1-0)}$	ALMA/IRAM	21.70″	0.33 ± 0.04	0.64 ± 0.07	0.66 ± 0.07
	IRAM/ALMA	10.85″	1.34 ± 0.19	0.67 ± 0.04	0.87 ± 0.09

Table 6: CO(2–1)/CO(1–0) velocity-integrated intensity ratios in NGC 3110 for the North, Center, and South pointings, measured after matching the data to the appropriate angular resolution.

On the other hand, the line intensity ratios involving the dense gas tracers play an important role in this study. The HCN-to-CO ratio has historically been used as an observational proxy for the dense molecular gas fraction and as an indicator of enhanced star-forming activity in galaxies. The HCO^+-to-CO ratio can be used in a similar way, since HCO^+ is also commonly employed as a tracer of dense molecular gas (Jiménez-Donaire et al. 2019). HNC likewise traces relatively dense gas, although its emission is generally weaker and its interpretation may be more sensitive to the excitation and chemical state of the gas (Israel 2023).

Ratios among the dense-gas tracers provide complementary information, since they compare molecular species that probe relatively dense gas but may respond differently to excitation and chemical conditions. The HCN/HCO^+ ratio is commonly used to compare the behavior of two of the main dense-gas tracers, and has also been proposed as an empirical indicator to distinguish between starburst- and AGN-dominated environments, although its interpretation is not unique. Ratios involving HNC are especially useful because this molecule can be more sensitive to the thermal and chemical state of the gas. In particular, HNC/HCN has been discussed in relation to changes in gas chemistry, excitation conditions, and the presence of relatively enhanced HNC emission (Israel 2023; Pérez-Beaupuits et al. 2007).

Table 7 shows the dense-to-bulk molecular gas ratios, $I_{\rm dense}/I_{\rm CO(1-0)}$, obtained by comparing HCN(1–0), HCO^+(1–0), and HNC(1–0) with the bulk molecular gas tracer CO(1–0). The detected ratios lie in the approximate range 0.02–0.05. HCN/CO and HCO^+/CO are measured in the three observed regions and show comparable values, particularly in the North and Center pointings. In the South region, HCN/CO reaches the highest value in the table, while HCO^+/CO remains slightly lower. The HNC/CO ratio is only detected in the Center pointing, where it is lower than the corresponding HCN/CO and HCO^+/CO ratios; in the North and South regions, only upper limits are obtained. Overall, only moderate variations are found among the three pointings, and these differences are comparable to the quoted uncertainties.

Ratio	North	Center	South
HCN(1–0)/CO(1–0)	0.040 ± 0.012	0.034 ± 0.005	0.047 ± 0.008
HCO^+(1–0)/CO(1–0)	0.042 ± 0.011	0.041 ± 0.005	0.037 ± 0.008
HNC(1–0)/CO(1–0)	< 0.022	0.020 ± 0.005	< 0.015

Table 7: Dense-to-bulk molecular gas intensity ratios in NGC 3110 for the North, Center, and South pointings. Dense gas data collected by IRAM and CO corresponding to ALMA maps, integrated over the same region. All ratios correspond to the $J = 1$–0 transitions.

For comparison with galaxy-integrated studies such as Gao & Solomon (2004), the measured line intensities can also be expressed as brightness-temperature line luminosities, L', in units of K km s^{-1} pc^2. This quantity measures the velocity-integrated brightness temperature over a physical area and is therefore the standard magnitude used in extragalactic molecular-line scaling relations. For a beam-averaged intensity, it can be written as

$$L'_{\rm line} = I_{\rm line}\, A_{\rm beam} = I_{\rm line}\, 1.133\, (\theta_{\rm HPBW}\, \mathcal{S})^2\,, \tag{3.5}$$

with $\mathcal{S} = 352$ pc arcsec^{-1} the physical scale adopted for NGC 3110 (Espada et al. 2018). The CO(1–0) luminosities listed in Tab. 8 are computed from the CO(1–0) ALMA map after simulating the same 27.83″ IRAM 30 m beam, so that the CO and dense-gas luminosities refer to the same physical area.

Line luminosity (10^7 K km s^{-1} pc^2)	North	Center	South
$L'_{\rm CO(1-0)}$	83 ± 4	266 ± 13	114 ± 5
$L'_{\rm HCN}$	3.5 ± 1.1	9.2 ± 1.2	5.4 ± 0.9
$L'_{\rm HCO^+}$	3.5 ± 1.0	11.4 ± 1.1	4.3 ± 0.9
$L'_{\rm HNC}$	< 2.0	5.3 ± 1.3	< 1.7

Table 8: Beam-matched brightness-temperature line luminosities for the three IRAM 30 m dense-gas pointings, in units of 10^7 K km s^{-1} pc^2. The CO(1–0) luminosity is derived from the ALMA CO(1–0) map after simulating the 27.83″ beam of the dense-gas observations. Upper limits are given for the non-detections of HNC(1–0) in the North and South pointings.

Since the CO(1–0) and dense-gas luminosities in Table 8 are computed over the same beam area, their luminosity ratios are equivalent to the corresponding beam-matched integrated-intensity ratios in Table 7. This allows a direct comparison with the classical $L'_{\rm HCN}/L'_{\rm CO}$ diagnostic used by Gao & Solomon (2004), while keeping in mind that the present values are regional beam luminosities rather than luminosities integrated over the whole galaxy.

Table 9 presents the line intensity ratios among the dense gas tracers in the three observed regions of NGC 3110. The HCN/HCO$^+$ ratio remains of order unity in all pointings. Ratios involving HNC are only directly measured in the Center region, where HNC/HCO$^+$ and HNC/HCN reach 0.47 ± 0.12 and 0.58 ± 0.16, respectively. In the North and South pointings, the non-detection of HNC leads to upper limits for these ratios. Overall, HCN and HCO$^+$ show comparable intensities across the three regions, whereas HNC appears weaker wherever it can be constrained.

Ratio	North	Center	South
HCN/HCO$^+$	1.0 ± 0.4	1.24 ± 0.21	0.79 ± 0.20
HNC/HCO$^+$	< 0.54	0.47 ± 0.12	< 0.42
HNC/HCN	< 0.55	0.58 ± 0.16	< 0.33

Table 9: Dense-gas tracer intensity ratios in NGC 3110 for the North, Center, and South pointings. All ratios correspond to the $J = 1$–0 transitions.

4 Analysis

4.1 Observational limitations and beam-averaged nature of results

The observations are limited in angular resolution by the IRAM 30 m telescope. Adopting the linear scale of 352 pc arcsec^{-1} used by Espada et al. (2018), the HPBW $= 27.83''$ beam of the dense-gas observations corresponds to a physical scale of ≈ 10 kpc[8]. This beam size is sufficiently large that the pointings do not isolate small regions associated with individual SSCs, but instead average over extended areas of the galaxy. This allows the different pointings to be compared in terms of large-scale, beam-averaged trends, but it also implies that local variations in the dense-gas tracers on the scale of individual SSCs may remain unresolved. In particular, the present data cannot spatially isolate the region of maximum star formation efficiency identified by Kawana et al. (2022) at the southern tip of the southern spiral arm, since that structure is strongly diluted within the much larger IRAM beam. Therefore, the results discussed below should be interpreted as beam-averaged regional diagnostics rather than as direct measurements of the molecular gas conditions on SSC scales.

On the other hand, the comparison between the IRAM observations and the ALMA moment 0 maps in Table 5 suggests that the central CO(2–1) emission, with an IRAM/ALMA flux ratio of 0.95 ± 0.10, is well recovered by ALMA. By contrast, in both galaxy arms, especially in the northern one, the extended CO(2–1) emission is underestimated by the interferometric map. These results are congruent with Espada et al. (2018), who estimated an overall flux recovery of $\sim$ 54% for their CO(2–1) interferometric observations, supporting the relevance of missing-flux effects in the interpretation of our data. They are also consistent with the fact that the CO emission in the arms of NGC 3110 is more extended and asymmetric. Considering this, the CO ratios should be interpreted with caution, particularly in the North region, because the outcome could be strongly influenced by interferometric filtering and missing-flux effects.

4.2 CO excitation across NGC 3110

Looking at Table 6, the Center pointing provides the most robust estimate of the CO(2–1)/CO(1–0) intensity ratio, $R_{21/10}$. The ratios obtained at both angular resolutions are mutually consistent, and the nearly complete CO(2–1) flux recovery further supports the reliability of this result. The central value, $R_{21/10} \sim 0.65$, indicates moderately excited molecular gas, compatible with optically thick but not fully thermalised emission, or with a mixture of molecular gas phases within the beam. Moreover, it is broadly coherent with the average value of ~ 0.8 reported by Kawana et al. (2022), taking into account the differences in angular resolution and methodology.

The southern region gives 0.66 ± 0.07 at the larger beam and increases to 0.87 ± 0.09 at the smaller one. This may suggest that the more compact CO emission in the southern arm is somewhat more excited, which agrees with Kawana et al. (2022). However, the South-to-Center difference is only $\sim$ 2.3% at $21.70''$ and

[8] $27.83'' \times 352$ pc arcsec$^{-1} \approx 9.8$ kpc.

$\sim$ 30% at 10.85″, so this behavior can be regarded as a suggestive, marginally significant trend, but not as conclusive evidence for enhanced CO excitation in the southern pointing.

Finally, the northern region requires the most cautious interpretation, as already noted above. Its ratio changes from 0.33 ± 0.04 at 21.70″ to 1.34 ± 0.19 at 10.85″. However, the strong discrepancy between the IRAM and ALMA CO(2–1) fluxes strongly limits any physical interpretation of this value.

4.3 Dense-to-bulk molecular gas ratios

In terms of absolute integrated intensities, Table 3 shows that all the molecular transitions observed in this work peak at the central pointing. This is particularly clear for CO(1–0), with $I_{\mathrm{CO}(1-0)} = 38.8 \pm 0.6$ K km s^{-1}, about four to five times higher than in the spiral-arm pointings. The dense-gas tracers also reach their maximum values in the center, with $I_{\mathrm{HCN}} = 0.85 \pm 0.11$ K km s^{-1} and $I_{\mathrm{HCO^+}} = 1.05 \pm 0.10$ K km s^{-1}. In addition, HNC(1–0) is securely detected only in the central region. This behavior indicates that the molecular line emission is strongly concentrated toward the center of NGC 3110, in agreement with the central molecular gas concentration reported by Espada et al. (2018) and Kawana et al. (2022).

However, the picture changes when the dense-gas tracers are normalized by CO(1–0). As shown in Table 7, neither HCN(1–0)/CO(1–0) nor HCO$^+$(1–0)/CO(1–0)[9] peaks clearly at the central pointing, with central values of $R_{\mathrm{HCN/CO}} = 0.034 \pm 0.005$ and $R_{\mathrm{HCO^+/CO}} = 0.041 \pm 0.005$. These values are comparable, within the uncertainties, to those measured in the spiral arms. In particular, HCN(1–0)/CO(1–0) is slightly higher in the South pointing, while HCO$^+$(1–0)/CO(1–0) remains very similar between the Center and North regions. Although the center shows the brightest beam-averaged molecular emission and the strongest CO-traced molecular reservoir, it does not show a clearly enhanced dense-to-bulk molecular gas ratio with respect to the spiral arms.

The luminosity calculation introduced in Table 8 provides the appropriate framework to compare these results with the dense-gas study of Gao & Solomon (2004). In that work, the brightness-temperature luminosity ratio $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$ was interpreted as an indicator of the dense molecular gas fraction and as a starburst diagnostic. Gao & Solomon found that normal spiral galaxies show relatively low values, typically in the range $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}} \simeq 0.02$–$0.05$, while all galaxies in their sample with $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}} > 0.06$ are LIRGs or ULIRGs.

In NGC 3110, the values of $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$ are 0.040 ± 0.012 in the North, 0.034 ± 0.005 in the Center, and 0.047 ± 0.008 in the South. These values do not change in comparison with the corresponding beam-matched intensity ratios in Table 7, since the CO(1–0) and dense-gas luminosities were computed over the same physical area. Therefore, the three pointings lie below the classical $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}} \simeq 0.06$ threshold associated with the most HCN-enhanced LIRG and ULIRG systems in the Gao & Solomon framework.

[9] From now on, this type of quotients will be defined as $R_{\mathrm{dense/CO}}$.

This comparison should be interpreted carefully. Gao & Solomon (2004) used galaxy-integrated luminosities, whereas the luminosities derived in this work are regional and hence do not represent the total molecular luminosity of NGC 3110. Nevertheless, they are useful to place the observed regions in the context of classical dense-gas scaling relations. In this sense, the South pointing shows the highest $L'_{\rm HCN}/L'_{\rm CO}$ value, but the enhancement is modest and does not place it in an extreme dense-gas regime.

Furthermore, in a more recent study, Israel (2023) argues that, in galaxy centers, $R_{\rm HCN/CO}$ should not be used as a direct proxy for the dense gas fraction, because the observed low-J line intensities and luminosities are affected by excitation, optical depth, abundance, and heating mechanisms. Israel reports average values close to $R_{\rm HCN/CO} \sim 0.046$, $R_{\rm HCO^+/CO} \sim 0.041$, and $R_{\rm HNC/CO} \sim 0.021$ for central molecular regions. The corresponding values in NGC 3110, $R_{\rm HCN/CO} \simeq 0.033$–$0.047$, $R_{\rm HCO^+/CO} \simeq 0.037$–$0.042$, and $R_{\rm HNC/CO} = 0.020 \pm 0.005$ in the Center pointing, are not extreme, but they are consistent with the range found in central molecular zones of nearby galaxies. This is noteworthy because even the spiral-arm pointings of NGC 3110 show dense-gas-tracer ratios comparable to those usually associated with more central or high-pressure environments.

This interpretation is reinforced by the comparison with the EMPIRE survey. Jiménez-Donaire et al. (2019) find average ratios of $R_{\rm HCN/CO} = 0.025$, $R_{\rm HCO^+/CO} = 0.018$, and $R_{\rm HNC/CO} = 0.011$ when disk and central regions are considered together. Their central pointings show somewhat higher values, with $R_{\rm HCN/CO} \simeq 0.030$–$0.034$, $R_{\rm HCO^+/CO} \simeq 0.024$–$0.025$, and $R_{\rm HNC/CO} \simeq 0.013$–$0.014$. Compared with these values, NGC 3110 shows dense-gas-tracer emission relative to CO that is higher than typical disk averages and closer to, or slightly above, the values usually found in central regions of nearby spiral galaxies.

The moderate variations of $R_{\rm HCN/CO}$ and $R_{\rm HCO^+/CO}$ across NGC 3110 are also relevant. $R_{\rm HCN/CO}$ reaches its maximum value in the South pointing, while $R_{\rm HCO^+/CO}$ reaches its lowest value in this region instead. If the southern arm showed a clear increase in the dense molecular gas fraction, a more coherent enhancement in both dense-gas tracers would be expected. This is particularly important because Israel (2023) finds that HCN and $\rm HCO^+$ have very similar intensities on average and trace the same molecular gas component, despite their different critical densities. Similarly, Jiménez-Donaire et al. (2019) find that $\rm HCO^+$ generally follows a similar behavior to HCN in nearby galaxy disks. Taking into account the uncertainties of each value, they lie in the same interval, following that despite the lack of simultaneous increase in both $R_{\rm HCO/CO}$ and $R_{\rm HCO^+/CO}$, there is a possible trend for a larger dense gas fraction in the South region.

The behavior of HNC(1–0)/CO(1–0) is more difficult to interpret because HNC is securely detected only in the Center pointing, while the North and South values are upper limits. The central value, $R_{\rm HNC/CO} = 0.020 \pm 0.005$, is lower than the HCN/CO and $\rm HCO^+/CO$ ratios, as expected since HNC is generally the faintest of the three dense-gas tracers. Jiménez-Donaire et al. (2019) also find that HNC/CO is typically lower than HCN/CO and $\rm HCO^+/CO$ in nearby galaxy disks. Therefore, the HNC result is coherent with the general weakness of this tracer, but the upper limits in the spiral arms prevent us from using it to establish a clear regional trend.

Overall, the dense-to-bulk ratios indicate that NGC 3110 has relatively high dense-gas-tracer emission compared with normal star-forming disks, but without showing a strong contrast between the observed pointings. The center dominates the absolute molecular emission, whereas the ratios normalized by CO remain similar within the uncertainties.

4.4 Dense-gas tracer ratios

The ratios between the dense-gas tracers provide complementary information to the dense-to-CO ratios discussed above. In particular, the HCN(1–0)/HCO^+(1–0) ratio, showed in Table 9, is useful to compare the behavior of two of the most commonly used dense-gas tracers. In NGC 3110, this ratio remains close to unity in the three pointings, with values of 1.0 ± 0.4, 1.24 ± 0.21, and 0.79 ± 0.20 for the North, Center, and South regions, respectively. Consequently, there is no evidence for a strong enhancement of HCN with respect to HCO^+ in any of the observed regions.

This result agrees well with the interpretation of Israel (2023). In that study, HCN and HCO^+ are found to have very similar intensities on average and to trace the same molecular gas component in nearby galaxy centers. The ratios measured in NGC 3110 are fully in accordance with this picture. The central value is slightly higher than unity, but it is not extreme and remains compatible with the typical range found in extragalactic dense-gas observations. As a result, the HCN/HCO^+ ratios do not suggest a strongly anomalous chemistry or excitation state in any of the observed pointings.

The HCN/HCO^+ ratio can also be discussed in the context of AGN and starburst diagnostics. Enhanced HCN/HCO^+ ratios have sometimes been associated with AGN-dominated environments, while starburst-dominated regions often show values closer to unity. However, this interpretation is not unique, and the ratio alone cannot be used as a definitive proof, since HCO^+ may appear depressed relative to HCN in nuclei dominated by an active galactic nucleus, resulting in higher HCN/HCO^+ values without requiring an excess of HCN emission. In light of the measurements in NGC 3110, the dense-gas tracer ratios do not provide evidence for an AGN-like molecular signature. In contrast, they suggest a star-formation-dominated molecular medium.

The HNC-related ratios require a more cautious interpretation. HNC(1–0) is securely detected only in the Center pointing, where the measured ratios are $R_{\mathrm{HNC/HCN}} = 0.58 \pm 0.16$ and $R_{\mathrm{HNC/HCO^+}} = 0.47 \pm 0.12$. These values indicate that HNC is weaker than both HCN and HCO^+ in the central region, but not unusually faint when compared with typical extragalactic measurements. The upper limits in the spiral arms, $R_{\mathrm{HNC/HCN}} < 0.55$ in the North and $R_{\mathrm{HNC/HCN}} < 0.33$ in the South, suggest that HNC is not enhanced with respect to HCN outside the center.

The interpretation of HNC/HCN should not be reduced to a simple temperature diagnostic. Pérez-Beaupuits et al. (2007) show that bright or weak HNC emission can be produced by several mechanisms, including hidden cold gas, ion-neutral chemistry, X-ray dominated chemistry, and mid-infrared pumping. Although the central value of 0.58 ± 0.16 indicates that HNC is weaker than HCN,

it does not allow a unique conclusion about the kinetic temperature or the dominant chemical regime of the dense gas. Multi-transition observations would be required to distinguish between excitation, chemistry, and radiative-transfer effects.

4.5 SSC surface density and molecular gas ratios

The SSC surface-density map provides an additional observational constraint to compare the molecular gas conditions with the distribution of recent clustered star formation. As shown in Fig. 7 and Table 2, the three IRAM 30 m pointings sample regions with significant K_s-selected SSC candidate populations. In this comparison, the relevant quantity is the CO-normalized SSC surface density, Σ_{SSC}, since it measures the number of SSC candidates per unit area covered by CO(1–0) emission within each IRAM aperture.

The Center pointing shows the highest value, with $\Sigma_{\mathrm{SSC}} = 5.37\ \mathrm{kpc}^{-2}$, followed by the North and South pointings, with $\Sigma_{\mathrm{SSC}} = 3.24\ \mathrm{kpc}^{-2}$ and $1.63\ \mathrm{kpc}^{-2}$, respectively. Given this circumstance, the center region combines the highest SSC surface density with the strongest molecular line emission and the largest dense-gas tracer luminosities. However, this maximum in Σ_{SSC} does not correspond to a maximum in the observed dense-to-bulk molecular gas luminosity ratios.

Pointing	N_{SSC}	Σ_{SSC} (kpc^{-2})	$L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$	$L'_{\mathrm{HCO^+}}/L'_{\mathrm{CO}}$	$R_{21/10}$ (10.85″)
North	66	3.24	0.040 ± 0.012	0.042 ± 0.011	1.34 ± 0.19
Center	274	5.37	0.034 ± 0.005	0.041 ± 0.005	0.67 ± 0.04
South	47	1.63	0.047 ± 0.008	0.037 ± 0.008	0.87 ± 0.09

Table 10: Comparison between the number of K_s-selected SSC candidates, the CO-normalized SSC surface density, Σ_{SSC}, and the main beam-matched molecular ratios discussed in the analysis. The luminosity ratios $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$ and $L'_{\mathrm{HCO^+}}/L'_{\mathrm{CO}}$ compare the dense-gas tracer luminosities with the CO(1–0) luminosity over the same 27.83″ beam. The North value of $R_{21/10}$ should be interpreted with particular caution because of missing-flux effects affecting the CO(2–1) emission in this region.

The comparison in Table 10 shows that Σ_{SSC} does not follow a simple monotonic trend with the observed dense-gas-tracer luminosity ratios. The Center pointing has already been discussed. On the other hand, the South pointing has the largest $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$ value, despite having the lowest Σ_{SSC} among the three pointings. Moreover, this southern enhancement is not accompanied by a simultaneous increase in $L'_{\mathrm{HCO^+}}/L'_{\mathrm{CO}}$.

However, the luminosity ratios should not be interpreted directly as mass fractions without considering the conversion factors. If the dense molecular gas mass and the bulk molecular gas mass are written as

$$M_{\mathrm{dense}} = \alpha_{\mathrm{HCN}} L'_{\mathrm{HCN}}, \qquad M_{\mathrm{mol}} = \alpha_{\mathrm{CO}} L'_{\mathrm{CO}}, \tag{4.1}$$

then the dense-gas mass fraction traced by HCN is

$$\frac{M_{\mathrm{dense}}}{M_{\mathrm{mol}}} = \frac{\alpha_{\mathrm{HCN}}}{\alpha_{\mathrm{CO}}} \frac{L'_{\mathrm{HCN}}}{L'_{\mathrm{CO}}}. \tag{4.2}$$

Thus, the observed ratio $L'_{\rm HCN}/L'_{\rm CO}$ is only proportional to the dense-gas mass fraction if $\alpha_{\rm HCN}$ and $\alpha_{\rm CO}$ are constant between regions. For NGC 3110, Kawana et al. (2022) estimated a typical value of $\alpha_{\rm CO} \simeq 1.7\ M_\odot\ (\mathrm{K\,km\,s^{-1}\,pc^2})^{-1}$, but it is uncertaint that this is constant across the galaxy. Therefore, if $\alpha_{\rm CO}$ is lower in the central starburst region than in the spiral arms, the true value of $M_{\rm dense}/M_{\rm mol}$ in the Center would be higher than suggested by $L'_{\rm HCN}/L'_{\rm CO}$ alone.

The uncertainty is even larger for $\alpha_{\rm HCN}$. Although HCN(1–0) is commonly used as a dense-gas tracer, the conversion from HCN luminosity to dense-gas mass is not universal and may depend on excitation, optical depth, abundance, density distribution and cloud dynamical state (Israel 2023). Consequently, part of the lack of a direct correspondence between $\Sigma_{\rm SSC}$ and $L'_{\rm HCN}/L'_{\rm CO}$ could be caused by regional variations in $\alpha_{\rm CO}$ and $\alpha_{\rm HCN}$, rather than by a real absence of connection between SSC formation and the dense-gas mass fraction.

The CO excitation ratio also does not produce a simple correspondence with $\Sigma_{\rm SSC}$. The South pointing shows a higher $R_{21/10}$ than the Center at the smaller beam, which is compatible with somewhat more excited compact molecular gas in the southern arm. However, the North value is strongly affected by missing-flux effects in the CO(2–1) map, and therefore should not be interpreted with the same physical confidence. Consequently, the present data suggest that the SSC surface density, the observed dense-to-bulk molecular gas luminosity ratios, and the CO excitation do not vary in a single coherent sequence across the three pointings.

This comparison is limited by the beam-averaged nature of the measurements. The values of $\Sigma_{\rm SSC}$ are computed within large $27.83''$ apertures, and the pointings partially overlap. In addition, the high-SFE region identified by Kawana et al. (2022) at the southern tip of the southern spiral arm is much more localized than the IRAM beam. Therefore, the SSC surface-density map supports the relevance of the selected pointings, but it also shows that compact SSC-forming regions may be diluted in the beam-averaged molecular ratios.

5 Conclusions

The object of study has been the molecular gas conditions in NGC 3110 through IRAM 30 m observations of CO(1–0), CO(2–1), HCN(1–0), $\mathrm{HCO^+}$(1–0), and HNC(1–0), complemented with ALMA CO(1–0) and CO(2–1) maps. The aim was to investigate whether the molecular gas properties measured in the center and spiral arms are related to the onset of intense star formation and to the formation of super star clusters in this interacting LIRG.

The clearest observational result is the strong concentration of molecular emission toward the central pointing. CO(1–0) reaches $38.8 \pm 0.6\ \mathrm{K\,km\,s^{-1}}$ in the Center, compared with 9.0 ± 0.5 and $8.2 \pm 0.4\ \mathrm{K\,km\,s^{-1}}$ in the North and South pointings, respectively. The dense-gas tracers follow the same trend: HCN(1–0) peaks at $0.85 \pm 0.11\ \mathrm{K\,km\,s^{-1}}$, $\mathrm{HCO^+}$(1–0) at $1.05 \pm 0.10\ \mathrm{K\,km\,s^{-1}}$, and HNC(1–0) is securely detected only in the Center, with $0.49 \pm 0.12\ \mathrm{K\,km\,s^{-1}}$. Consequently, the IRAM data confirm that the central region contains the brightest beam-averaged molecular component among the three observed regions, in agreement with the central molecular concentration previously reported in NGC 3110.

The CO excitation analysis gives its most reliable result also in the central pointing, where the CO(2–1)/CO(1–0) ratio is compatible at the two angular resolutions considered: $R_{21/10} = 0.64 \pm 0.07$ at 21.70″ and 0.67 ± 0.04 at 10.85″. These values indicate moderately excited molecular gas, compatible with optically thick but not fully thermalised emission or with a mixture of molecular gas phases inside the beam. In the South pointing, the ratio increases from 0.66 ± 0.07 at 21.70″ to 0.87 ± 0.09 at 10.85″, suggesting a somewhat more compact and more excited CO component there. The North value at 10.85″, 1.34 ± 0.19, is not considered a reliable excitation measurement because the ALMA CO(2–1) emission is strongly affected by missing flux in that region.

The dense-to-bulk molecular gas ratios do not follow the same ordering as the molecular line intensities. Although the Center dominates the absolute CO, HCN, HCO^+, and HNC emission, it does not show the highest HCN-to-CO ratio, $R_{HCN/CO}$. The measured values are $R_{HCN/CO} = 0.040 \pm 0.012$, 0.034 ± 0.005, and 0.047 ± 0.008 in the North, Center, and South pointings, respectively. Thus, the South pointing has the highest nominal $R_{HCN/CO}$ value, but $R_{HCO^+/CO}$ does not show the same behavior, with values of 0.042 ± 0.011, 0.041 ± 0.005, and 0.037 ± 0.008 in the North, Center, and South regions. Hence, the southern pointing may show a mild enhancement in HCN relative to CO, but the lack of a simultaneous increase in $R_{HCO^+/CO}$ means that this cannot be interpreted as strong evidence for a larger dense-gas fraction.

The luminosity ratios lead to the same conclusion. The regional values of L'_{HCN}/L'_{CO} are 0.040 ± 0.012, 0.034 ± 0.005, and 0.047 ± 0.008 in the North, Center, and South pointings, respectively. All three values remain below the classical $L'_{HCN}/L'_{CO} \simeq 0.06$ threshold associated with the most HCN-enhanced LIRGs and ULIRGs in the Gao & Solomon framework. Therefore, NGC 3110 shows significant dense-gas-tracer emission, but none of the three observed regions reaches an extreme HCN luminosity excess at the spatial scale considered.

Considering gas mass fractions instead of luminosity ratios, the conversion factors have to be taken into account. L'_{HCN}/L'_{CO} traces the dense-gas mass fraction only if the conversion factors are constant between regions. This is especially relevant in NGC 3110 because Kawana et al. (2022) adopt $\alpha_{CO} \simeq 1.7\ M_\odot\ K^{-1}\ km^{-1}\ s\ pc^{-2}$, lower than the commonly adopted reference value $\alpha_{CO} \simeq 4.36\ M_\odot\ K^{-1}\ km^{-1}\ s\ pc^{-2}$ reviewed by Bolatto et al. (2013). Moreover, α_{HCN} is even less constrained and can depend on excitation, opacity, abundance, and cloud structure. Consequently, the observed luminosity ratios provide robust empirical comparisons between regions, but the corresponding dense-gas mass fractions remain uncertain.

Although the ratios are not extreme, they are high compared with normal star-forming disks. The values $R_{HCN/CO} \simeq 0.033$–0.047, $R_{HCO^+/CO} \simeq 0.037$–$0.042$, and the central $R_{HNC/CO} = 0.020 \pm 0.005$ are comparable to the central molecular-zone averages reported by Israel (2023), and they are above the disk-averaged ratios found in the EMPIRE survey, especially for HCO^+/CO and HNC/CO (Jiménez-Donaire et al. 2019). This indicates that even the spiral-arm pointings of NGC 3110 show dense-gas-tracer emission more similar to central or high-pressure environments than to ordinary galaxy disks.

The ratios among the dense-gas tracers do not provide evidence for an AGN-like molecular signature. The HCN/HCO^+ ratio remains close to unity, with values of 1.0 ± 0.4, 1.24 ± 0.21, and 0.79 ± 0.20 in the North, Center, and South pointings, respectively. These values show that HCN and HCO^+ behave similarly across the three regions. The central value is slightly above unity, but it is not large enough to indicate a strong HCN enhancement. Therefore, the molecular line ratios are more in agreement with a star-formation-dominated medium than with an AGN-dominated chemistry or excitation state.

HNC provides useful information mainly in the central region. It is securely detected only in the Center, where $R_{\mathrm{HNC/HCN}} = 0.58 \pm 0.16$ and $R_{\mathrm{HNC/HCO^+}} = 0.47 \pm 0.12$. These ratios confirm that HNC is weaker than both HCN and HCO^+, as commonly found in nearby galaxies (Pérez-Beaupuits et al. 2007; Israel 2023). In the spiral arms, the upper limits $R_{\mathrm{HNC/HCN}} < 0.55$ in the North and $R_{\mathrm{HNC/HCN}} < 0.33$ in the South do not indicate any HNC enhancement outside the center. As a result, HNC does not reveal a distinct HNC-bright component in NGC 3110.

The comparison with the SSC surface-density map shows that the projected SSC distribution does not scale monotonically with the observed dense-gas-tracer ratios. The Center contains the largest number of SSC candidates, $N_{\mathrm{SSC}} = 274$, and the highest CO-normalized SSC surface density, $\Sigma_{\mathrm{SSC}} = 5.37\ \mathrm{kpc}^{-2}$. The North and South pointings contain $N_{\mathrm{SSC}} = 66$ and 47, with $\Sigma_{\mathrm{SSC}} = 3.24$ and $1.63\ \mathrm{kpc}^{-2}$, respectively. However, the Center does not have the highest HCN/CO or $L'_{\mathrm{HCN}}/L'_{\mathrm{CO}}$ ratio, while the South pointing has the highest nominal HCN/CO value despite having the lowest Σ_{SSC}. Taking this into account, at the ~ 10 kpc scale of the dense-gas IRAM beam, the relation between clustered star formation and dense-gas-tracer emission is not one-to-one.

Overall, NGC 3110 is not a system in which the onset of starburst activity can be explained by a single enhanced molecular ratio. The central region dominates the molecular reservoir and the SSC population, while the southern arm shows the highest HCN/CO ratio and a possible increase in compact CO excitation. Taken together, these results support a mixed scenario in which molecular gas concentration, moderate excitation variations, dense-gas-tracer emission, and interaction-driven gas compression all contribute to the star-forming activity. This agrees with the picture proposed by Espada et al. (2018) and Kawana et al. (2022), in which the minor interaction with MCG–01–26–013 promotes gas compression and fragmentation not only in the nucleus, but also along the spiral arms.

The main limitation of this work is the angular resolution of the dense-gas observations. The $27.83''$ IRAM beam corresponds to ~ 10 kpc at the distance of NGC 3110, so the measurements average over regions much larger than individual SSC-forming sites. The present results are therefore meaningful as regional, beam-averaged diagnostics, but they cannot isolate the compact high-SFE region identified by Kawana et al. (2022) at the southern tip of the southern arm. Higher-resolution observations of HCN, HCO^+, and HNC would be required to test whether compact dense-gas concentrations are diluted within the present single-dish beam.

Summary

1. The Center pointing dominates the molecular emission. CO(1–0) reaches 38.8 ± 0.6 K km s^{-1}, compared with 9.0 ± 0.5 in the North and 8.2 ± 0.4 K km s^{-1} in the South. HCN and HCO$^+$ also peak in the Center, with 0.85 ± 0.11 and 1.05 ± 0.10 K km s^{-1}, respectively.

2. The most reliable CO excitation result is obtained in the Center, with $R_{21/10} = 0.64 \pm 0.07$ at $21.70''$ and 0.67 ± 0.04 at $10.85''$. The South pointing increases from 0.66 ± 0.07 to 0.87 ± 0.09 at the smaller beam, suggesting a somewhat more compact and excited CO component in the southern arm.

3. The dense-to-bulk ratios do not peak in the Center. The HCN/CO values are 0.040 ± 0.012, 0.034 ± 0.005, and 0.047 ± 0.008 in the North, Center, and South pointings. The South has the highest nominal HCN/CO ratio, but HCO$^+$/CO does not increase there, with values of 0.042 ± 0.011, 0.041 ± 0.005, and 0.037 ± 0.008 for the respective pointings.

4. The luminosity ratios $L'_{\rm HCN}/L'_{\rm CO} = 0.040 \pm 0.012$, 0.034 ± 0.005, and 0.047 ± 0.008 all remain below the classical Gao & Solomon threshold of ~ 0.06. Hence, NGC 3110 does not show an extreme HCN luminosity excess. However, these ratios are not direct mass fractions, since the inferred $M_{\rm dense}/M_{\rm mol}$ depends on $\alpha_{\rm HCN}/\alpha_{\rm CO}$.

5. The measured dense-gas-tracer ratios are nevertheless elevated compared with normal star-forming disks. Even the spiral-arm pointings show HCN/CO and HCO$^+$/CO values closer to central or high-pressure molecular environments than to typical disk averages.

6. The HCN/HCO$^+$ ratio remains close to unity: 1.0 ± 0.4 in the North, 1.24 ± 0.21 in the Center, and 0.79 ± 0.20 in the South. These values do not provide evidence for an AGN-like molecular signature in the observed pointings.

7. HNC is weaker than HCN and HCO$^+$ and is securely detected only in the Center. The central ratios are $R_{\rm HNC/HCN} = 0.58 \pm 0.16$ and $R_{\rm HNC/HCO^+} = 0.47 \pm 0.12$, while the spiral arms only provide upper limits.

8. The SSC comparison shows that $\Sigma_{\rm SSC}$ is highest in the Center, with 5.37 kpc^{-2}, followed by the North and South with 3.24 and 1.63 kpc^{-2}. However, the Center does not have the highest HCN/CO ratio, and the South has the highest HCN/CO despite the lowest $\Sigma_{\rm SSC}$. Therefore, the relation between SSC formation and dense-gas-tracer ratios is not one-to-one at the IRAM 30 m resolution.

9. The main physical conclusion is that SSC formation in NGC 3110 is not governed by a single molecular ratio. The data support a scenario in which the central molecular concentration, moderate excitation variations, dense-gas-tracer emission, and interaction-driven gas compression all contribute to the ongoing starburst activity. Higher-resolution HCN, HCO$^+$, and HNC observations are needed to isolate the compact SSC-forming regions.

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